

DA42-VI HANDBOOK



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SECTION 1 AIRCRAFT MODIFICATION LIST

Modification	Source	Installed	
Maximum Landing Mass 1999 kg	MÄM 42-659	☒ yes	☐ no
Maximum Take-Off Weight 1999 kg and Maximum Zero Fuel Mass 1835 kg	MÄM 42-678	☒ yes	☐ no
Engine Software VC33_2_05_19*	MÄM 42-938	☒ yes	☐ no
Garmin Hard- and Software Upgrade I (Garmin G1000 NXi)	MÄM 42-978	☒ yes	☐ no
Garmin Hard- and Software Upgrade II (Garmin G1000 NXi)	MÄM 42-1072	☒ yes	☐ no
Ice Protection System	OÄM 42-053	☐ yes	☒ no
Oxygen System	OÄM 42-055	☐ yes	☒ no
Auxiliary Fuel Tanks	OÄM 42-056	☒ yes	☐ no
Front Seats with Adjustable Backrest	OÄM 42-067	☐ yes	☒ no

Modification	Source	Installed	
Garmin GWX 70 Weather Radar	OÄM 42-273	☐ yes	☒ no
Gear Warning Mute Function	OÄM 42-288	☒ yes	☐ no
Emergency Egress Hammer	OÄM 42-304	☒ yes	☐ no

* Or later approved software

Modification	Source	Installed	
Electrical Rudder Pedal Adjustment	OÄM 42-070	☐ yes	☒ no
Sun Visors	OÄM 42-101 OR OÄM 42-142	☒ yes	☐ no
Garmin GWX 68 Weather Radar	OÄM 42-119	☐ yes	☒ no
Ice Protection System	OÄM 42-160	☐ yes	☒ no
Ice Protection System with TKS Tank in Rear Fuselage	OÄM 42-160 AND OÄM 42-203	☐ yes	☒ no
Garmin G1000, SBAS Operation	OÄM 42-179	☒ yes	☐ no
Removal of Variable Elevator Stop	OÄM 42-199	☐ yes	☒ no
Emergency Axe	OÄM 42-205	☐ yes	☒ no
Short Baggage Extension	OÄM 42-207	☐ yes	☒ no
Electronic Stability and Protection (ESP)	OÄM 42-209	☒ yes	☐ no
Mission Alternator	OÄM 42-218	☐ yes	☒ no
Removal of Unfeathering Accumulator	OÄM 42-224	☐ yes	☒ no
Diesel Operation	OÄM 42-251	☐ yes	☒ no
Garmin GTX 3X5R Transponders for ADS-B	OÄM 42-258	☒ yes	☐ no
Front Seats with Adjustable Backrest - Hydrolok	OÄM 42-259	☒ yes	☐ no
Maximum Take-Off Weight 2001 kg / 4411 lb	OÄM 42-260	☐ yes	☒ no
Mid Continent MD302 Standby Attitude Module	OÄM 42-270	☐ yes	☒ no

SECTION 2 DIMENSIONS

Overall Dimensions

Span	: 13.42 m	44 ft
	: 13.55 m	44.5 ft including ACL
Length	: 8.56 m	28 ft 1 in
Height	: 2.49 m	8 ft 2 in

Wing

Airfoil	: Wortmann FX 63-137/20 - W4	
Wing Area	: 16.29 m ²	175.3 sq.ft.
Mean aerodynamic chord	: 1.271 m	4 ft 2 in
Aspect ratio	: 11.06	
Dihedral	: 5°	
Leading edge sweep	: 1°	

Aileron

Area (total, left + right)	: 0.66 m ²	7.1 sq.ft.
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Wing Flaps

Area (total, left + right)	: 2.18 m ²	23.5 sq.ft.
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Horizontal Tail

Area	: 2.35 m ²	25.3 sq.ft.
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Elevator area	: 0.66 m ²	7.1 sq.ft.
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Angle of incidence	: -1.1° relative to longitudinal axis of airplane	
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Vertical Tail

Area	: 2.43 m ²	26.2 sq.ft.
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Rudder area	: 0.78 m ²	8.4 sq.ft.
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Landing Gear

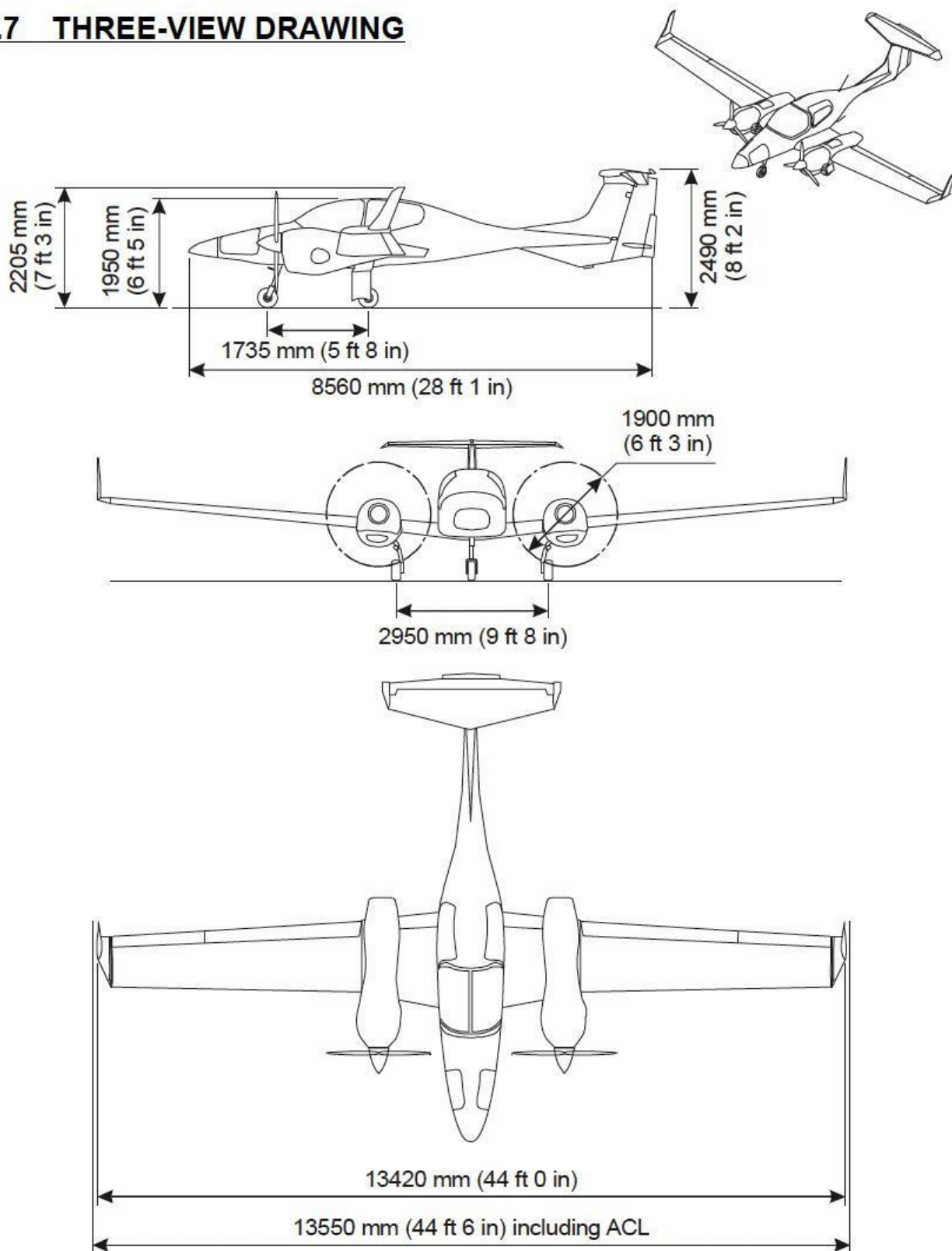
Track	: 2.95 m (9 ft 8 in)
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Wheelbase	: 1.735 m (5 ft 8 in)
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Nose wheel	: 5.00-5, for details refer to the AMM
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Main wheel	: 15x6.0-6, for details refer to the AMM
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1.7 THREE-VIEW DRAWING



SECTION 3 OPERATING LIMITATIONS

SUB-SECTION 3-1: AIRSPEED LIMITATIONS

V_O:

- Above 1800KG: 122KIAS
- Between 1700 and 1800KG: 119KIAS
- Below 1700KG: 112KIAS

V_{FE}:

- APP: 133KIAS

- LDG: 113KIAS

V_{LO}: Maximum gear extend/retract speed.

- Extension: V_{LOE} 188KIAS - Retraction: V_{LOR} 153KIAS

V_{LE}: Maximum speed with gear extended.

- 188KIAS

V_{MCA}: Minimum speed to maintain with OEI.

- APP: 68KIAS
- LDG: 71KIAS

V_{NO}: Yellow range speed.

- 151KIAS

V_{NE}: Red range speed.

- 188KIAS

V_{SSE}: Minimum Control Speed for Schooling: no value given...

V_R: Rotation speed.

- Flaps UP: Normal T/O: 76KIAS.



- Flaps APP: Short Field T/O: 74KIAS.

V_S : Stalling speed.

- V_{S0} : Given config: Flaps LDG, Gear DN: 64KIAS
- V_S : Given config: Flaps APP, Gear DN: 68KIAS
- V_S : Given config: Flaps UP, Gear UP: 72KIAS

V_X : Best Angle of Climb speed.

- Flaps UP: N/A.
- Flaps APP: 77KIAS.

V_Y : Best Rate of Climb speed.

- Flaps UP: 92KIAS. - Flaps APP: 85KIAS.

V_{YSE} : Best rate of climb speed with OEI.

- Also known as BLUE LINE speed.
- 85KIAS.



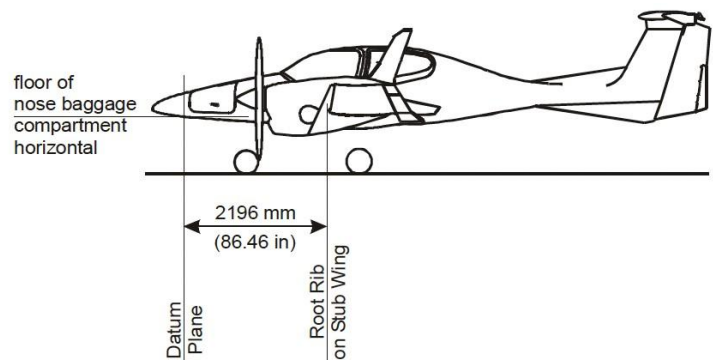
V_{REF} : Approach speed.

- Flaps UP: 92KIAS. In ice: 97KIAS. - Flaps APP: 88KIAS. In ice: 93KIAS.
- Flaps LDG: 86KIAS. In ice: PROHIBITED.

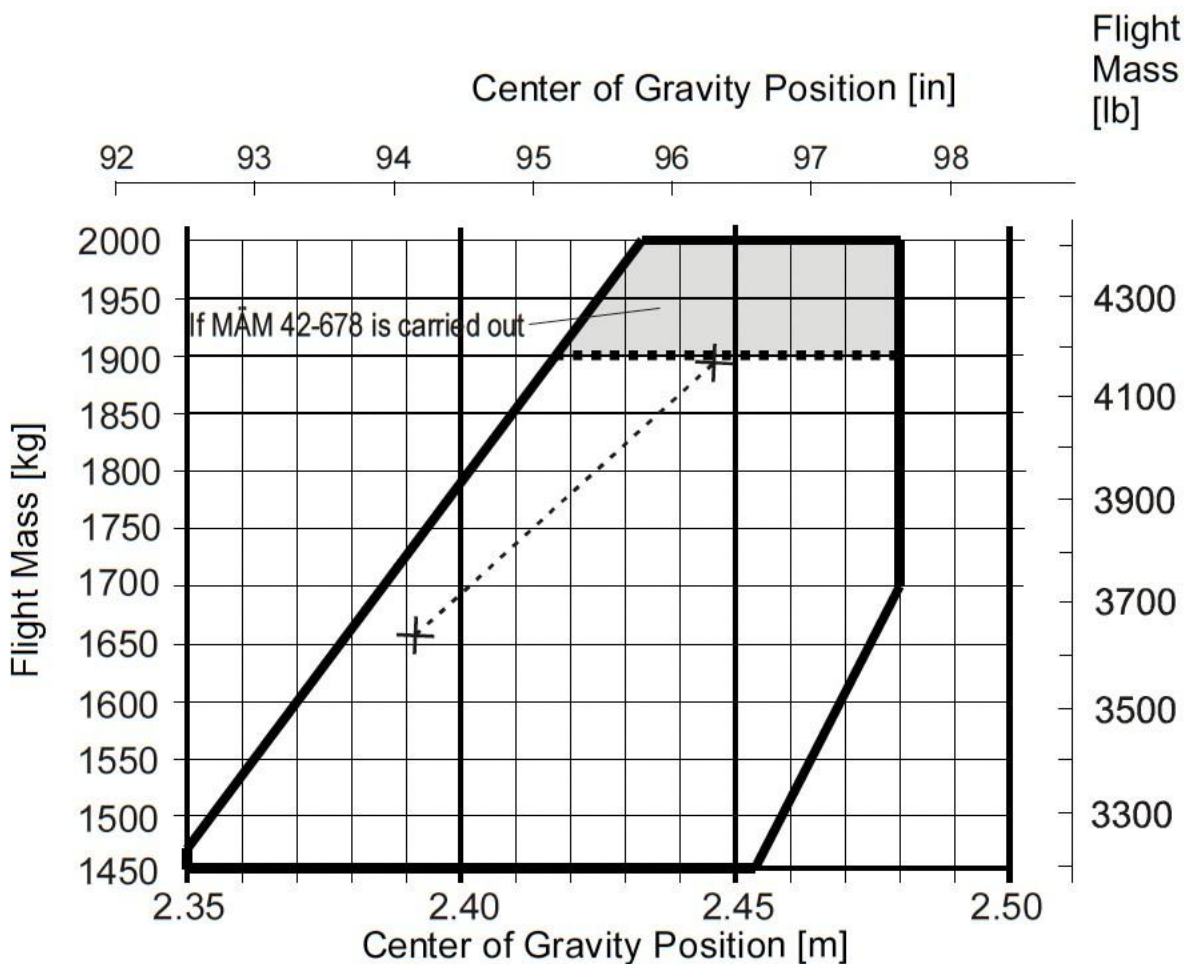
SUB-SECTION 3-2: MASS AND BALANCE LIMITATIONS:

Datum Plane:

The Datum Plane is located 2.196 meters (86.46 in) forward of the most forward point of the root rib on the stub wing.



Permissible CG range:



Maximum Structural Masses:

- MTOW: 1999KG
- MLW: 1999KG
- MZFW: 1835KG
- EMPTY MASS: 1450KG

Maximum Baggage Masses:

- Nose Compartment: 30KG
- Cockpit: 45KG
- Rear Extension: 18KG

Note: Max baggage mass in cockpit and rear extension combined 45KG.

For detailed M&B report, cf. "BALANCE SHEET DA42VI.xls"

SUB-SECTION 3-3: SYSTEM LIMITATIONS:

SUB-SECTION 3-3-1: POWER PLANT LIMITATIONS:

RPM Limitations:

- Max. T/O: 2300 RPM
- Max. CONT: 2300 RPM
- Max. ovspd: 2500 RPM

Engine Power:

- Max. T/O: 100% LOAD (max. 5mn) 123.5 kW
- Max. CONT: 92% LOAD 114 kW

Oil TEMP/PRESS/QTY:

- Temperature:
 - Min: -30°C
 - Max: 140°C

○ Norm. range: 50 – 135°C

- Pressure:

○ Min: Below 1500 RPM: 0.9 bar
Above 1500 RPM: 2.5 bar

○ Max: 6.5 bar

○ Norm. range: 2.5 – 6 bar

- Quantity:

○ Min: 5.0 L

○ Max: 7.0 L

○ Max. consump. 0.1L/h

Gearbox TEMP:

- Min (at startup): -30°C

- Min (full load): 35°C

- Max: 120°C

Coolant TEMP:

- Min (at startup): -30°C

- Min (full load): 60°C

- Max: 105°C

Fuel TEMP:

- Min: 4 bar

- Max: 7 bar

Restart Altitudes:

- For immediate: 15000ft
- OAT above -5°C: 10000ft (max ENG OFF time 10mn)

Restart Airspeeds:

- Starter assisted: max 100KIAS or airspeed for stationary prop, whichever is lower.
- Windmilling prop: 125-145KIAS

Note: no intentional IFSD below 3000ft AGL and above 10000ft PA.

SUB-SECTION 3-3-2: AIRFRAME LIMITATIONS:

Approved Maneuvers:

- All normal flight maneuvers.
- Stalling (except dynamic stalling)
- Lazy Eights, chandelles, steep turns all with a max of 60° of bank.
- Spinning, bank > 60°, stalling with OEI **prohibited**. - Intentional negative-g maneuvers **prohibited**.

Load Factor Limits:

- At V_O : $-1.52 < g < 3.8$
- At V_{NE} : *idem*
- Flaps APP/LDG: $0 < g < 2.0$

SUB-SECTION 3-3-3: AUTOPILOT LIMITATIONS:

- AP use during T/O, LDG and with OEI **prohibited**.
- Max speed 180KIAS - Min speed 90KIAS - Min. Alt.:

- DEP: 200ft AGL ○ ARR: 200ft AGL
- CRZ, CLB, maneuvering: 800ft AGL

SECTION 4 GENERAL RECOMMENDATIONS

Crossfeed Use:

- Max X-FEED test duration during taxi 30 sec.
- Simultaneous use of LH/RH X-FEED otherwise **prohibited.**
- T/O with X-FEED is **prohibited.**
- Simultaneous use of X-FEED and FUEL PUMPS is **prohibited.**

IFSD:

- “Dead foot, dead engine” in VMC.
- Engine instruments + ball in IMC.
- Feathering and power reduction on the “good engine” significantly reduce yaw momentum, adjust rudder trim accordingly. -